# Software Engineering Model Practice Paper

Software Engineering - Demo paper - 2025 edition - 50 marks

> *Self-created practice paper for NoteNest, not an official university paper.*

## Section A - Short answers (5 x 4 marks)

**Q1.** List the phases of the SDLC. **[4 marks]**

**Q2.** Differentiate between functional and non-functional requirements with two examples each. **[4 marks]**

**Q3.** What is coupling? Why is low coupling desirable? **[4 marks]**

**Q4.** Differentiate between black-box and white-box testing. **[4 marks]**

**Q5.** Define technical debt. **[4 marks]**

## Section B - Long answers (2 x 15 marks)

**Q6.** Compare Waterfall and Agile. Describe a project for which each is a better fit. **[15 marks]**

**Q7.** A function has 3 if statements and 1 while loop. Compute its cyclomatic complexity and propose test cases for boundary values of a field that accepts 1 to 100. **[15 marks]**

## Answer hints

- Q7: decision points = 4, so V(G) = 4 + 1 = 5.
- Q7: boundary tests for 1..100 are 0, 1, 2, 99, 100, 101.